
Towards Scalable Reachability: Globally Optimal Reachable Sets via Viscous Hamilton-Jacobi PDEs

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Computing safe reachable sets in verification settings for high-dimensional dy-
2 namic systems, parameterized by the viscous Hamilton-Jacobi (HJ) equation, in
3 general involves grid discretization of the phase space to achieve a numerically
4 consistent solution with respect to the original non-viscous problem. This presents
5 a two-fold problem: discretization implies exponential memory requirements; and,
6 computational costs increase as problem dimensions multiply. While parallel com-
7 puting can reduce the impact of the latter, it does not address the grid discretization
8 exponential memory costs of the former. Leveraging a recent connection between
9 HJ value functions, proximity operators, and the Moreau envelopes of HJ PDEs,
10 we cast the recovery of the safe set as an adaptive gradient descent optimization of
11 the Moreau envelope of the reachable HJI variational PDE. Our scheme provides
12 a storage-free means for resolving safe sets in a computationally time-efficient
13 manner without dealing with the exponential memory costs of grid discretization.

14 1 Introduction

15 This paper takes a further step towards improved computational schemes to high-dimensional prob-
16 lems described by time-dependent Hamilton-Jacobi (HJ) partial differential equations (PDEs) Evans
17 [2022]. These PDEs are increasingly employed in the safety verification Hobbs et al. [2023] of
18 cyberphysical systems. We work within a viscous HJ reachability ramification Lygeros [2004],
19 Mitchell et al. [2005], Molu [2024, 2025] and establish our results with globally optimal solutions to
20 reachability problems in nonlinear settings. In emphasis, we consider first-order nonlinear scalar HJ
21 PDEs of the form

$$\partial_t \mathbf{v} + \mathbf{H}(t; \mathbf{x}, \partial_x \mathbf{v}) = 0 \text{ in } \mathbb{R}^n \times (0, T], \quad \mathbf{v}(0; \mathbf{x}) = \mathbf{g}(\mathbf{x}) \text{ on } \mathbb{R}^n \times \{t = 0\} \quad (\text{HJ-IVP})$$

22 where $\partial_t \mathbf{v}$ denotes the partial derivative of the vector field $\mathbf{v}(\mathbf{x}, t)$ with respect to time, t ; the
23 Hamiltonian $\mathbf{H}(\mathbf{x}, t)$ is continuously defined on $\mathbb{R}^n$; $\mathbf{g}(\mathbf{x})$ is bounded and uniformly continuous
24 (BUC) in $\mathbb{R}^n$; and $\partial_x \mathbf{v}$ is the spatial gradient of $\mathbf{v}(\mathbf{x}, t)$.

25 While a unique C^2 solution $\mathbf{v}(\mathbf{x}, t)$ exists uniformly on $0 \leq t < T$ to (HJ-IVP) for which
26 $\lim_{t \rightarrow T} \mathbf{v}(\mathbf{x}, t)$ (when $\mathbf{H}$ and $\mathbf{g}$ are smooth and $\mathbf{v}_0(\mathbf{x}, t)$ is compactly supported), $\mathbf{H}(\mathbf{x}, t)$ lacks
27 continuous derivatives in this limit so that $\partial_x \mathbf{v}$ *vanishes* at $t = T$. However, one may find Lipschitz
28 continuous functions $\mathbf{v}(\mathbf{x}, t)$ on $\mathbb{R}^n \times (0, T]$ which are smooth *almost everywhere* (a.e.) and sat-
29 isfy (HJ-IVP) in a “generalized sense”. Suppose that we fix a viscosity term $\delta > 0$, Crandall and
30 Lions Crandall and Lions [1984] showed that $\mathbf{v}^\delta$ is the *vanishing viscosity* solution of

$$\partial_t \mathbf{v}^\delta + \mathbf{H}(t; \mathbf{x}, \partial_x \mathbf{v}^\delta) = \frac{\delta}{2} \Delta \mathbf{v}^\delta \text{ in } \mathbb{R}^n \times (0, T]; \quad \mathbf{v}^\delta(0; \mathbf{x}) = \mathbf{g}(\mathbf{x}) \text{ on } \mathbb{R}^n \times \{t = 0\} \quad (\text{HJ-Visc})$$

31 with Laplacian Δv^δ . For a constant $k > 0$, uniform convergence of $v^\delta \rightarrow v$ exists in the limit
 32 $\delta \rightarrow 0^+$, i.e.

$$\sup_{t \in (0, T)} \sup_{x \in \mathbb{R}^n} |v(t; x) - v^\delta(t; x)| \leq k\sqrt{\delta}. \quad (1)$$

33 In inf-sup or sup-inf optimal control problems Lygeros [2004], the Hamiltonian of (HJ-Visc) is
 34 related to the *backward* reachable set Mitchell et al. [2005] of a dynamical system. Reachable
 35 systems are those where one can compute all states that can be reached from an initial condition in a
 36 *finite number of steps*. Popular numerically consistent optimal solutions Crandall and Lions [1984],
 37 Crandall and Majda [1980a] to (HJ-Visc) utilize problem space grid discretization Osher and Fedkiw
 38 [2004], Mitchell et al. [2005]. However, these exact an exponential memory storage — making
 39 their practical applications on large problems limited. Recent efforts Molu [2024, 2025] leveraged
 40 single-instruction, multiple threads (SIMT) with modern GPUs to accelerate computation on a host
 41 of practical problems — showing improved spatial scalability with up to 76 – 92% computational
 42 improvement on higher-dimensional problems. However, this still employed grid-storage of the
 43 value function; as problem dimensions increase, the need for more memory renders such approaches
 44 unattractive. This work thus tack another thread, jettisoning grid-discretization methods for sampling-
 45 based proximal operators in a statistical moving average context. This resolves the global extrema
 46 of the Moreau envelopes of HJ cost functions, which can be used to recover the reachable set in
 47 nonconvex optimization settings Heaton et al. [2024], Crandall and Lions [2023], Chaudhari et al.
 48 [2018, 2019].

49 The Cole-Hopf transformation Evans [2022], Chaudhari et al. [2018] employed in PDE analysis relates
 50 the heat equation’s solution to that of (HJ-Visc). The essence of this work involves (i) first, leveraging
 51 the Cole-Hopf transformation to linearize the nonlinear HJ equation (HJ-Visc) to a heat equation.
 52 (ii) The computation of the spatial gradients of v^δ is a computationally intensive one in numerical
 53 HJ settings that typically employs upwinding schemes Osher and Fedkiw [2004], essentially non-
 54 oscillatory total variation diminishing Runge-Kutta Osher and Shu [1991] and Courants-Friedrichs-
 55 Lax adaptive time step weighting to recover a smooth solution to $\partial_x v^\delta$. (iii) We therefore devise a
 56 proximal operator equivalent to ii that enables a sampling-based evaluation of $\partial_x v^\delta$ in a stochastic and
 57 adaptive Moreau gradient descent algorithmic scheme under two inputs in a differential game setting.
 58 (iv) The solution v^δ to (HJ-Visc) is then recovered as a logarithm of the solution to the surrogate
 59 heat equation (devised via the Cole-Hopf transformation i). (v) Recovering the *robustly controlled*
 60 *backward reachable set or tube* (RCBRT) Mitchell [2020] in the terminal Hamilton-Jacobi-Isaacs
 61 equation Isaacs [1999], Mitchell et al. [2005], Molu [2024, 2025] of a typical reachability problem is
 62 thereafter turned into a lookup search problem for where the states fulfill the RCBRT condition.

63 The rest of this paper is structured as follows: §2, sets the background for this work. Section 3
 64 describes the theoretical and algorithmic machinery that sets the numerical results of section 4 in
 65 order. The paper is concluded in §5.

66 2 Abridged Background

67 We first introduce the notations that are commonly throughout this article. Reachable sets within
 68 the context of two person games Isaacs [1999] and their accompanying “viscous” terminal HJI
 69 PDE Evans and Souganidis [1984a] are then introduced. This is followed by the HJ-Isaacs PDEs
 70 commonly used to characterize reachable sets. A complete background that sets this work in context
 71 is provided in Appendix A.

72 2.1 Notations and Terminologies

73 Conventions: Upper-case and lower-case bold font Roman letters are matrices and vectors, respec-
 74 tively; calligraphic letters are sets. Exception: the subdifferential *set* of a BUC function, $\ell(x)$, is
 75 denoted ∂ . Time variables e.g. t_0, t, τ, T will always be real positive numbers. The n -dimensional
 76 Euclidean space is $\mathbb{R}^n$; the set of natural numbers is $\mathbb{N}$. The natural log of T is denoted $\log T$. The
 77 dot product is $\langle \cdot, \cdot \rangle$. For a function $f(x(t), y(t))$ with every argument parameterized by t , we simply
 78 write $f(t; x, y)$. We represent the real-valued Hilbert space with $\mathcal{H}$; denote $2^{\mathcal{H}}$ by its power set, i.e.
 79 the family of all subsets of $\mathcal{H}$. Associated with $\mathcal{H}$ is the maximally monotone set-valued operator,
 80 $Q : \mathcal{H} \rightarrow 2^{\mathcal{H}}$. The zeroes of Q are denoted $\text{zer } Q = \{x \in \mathcal{H} \mid 0 \in Qx\}$. The set of global

81 minimizers of $\ell(\mathbf{x})$ is denoted $\text{Argmin } \ell(\mathbf{x})$. It is written as $\text{argmin } \ell(\mathbf{x})$ when the set is a singleton.
 82 The set of fixed point of operator $S : \mathcal{H} \rightarrow \mathcal{H}$ is $\text{Fix } S = \{\mathbf{x} \in H \mid S\mathbf{x} = \mathbf{x}\}$.

83 The unique solution $\mathbf{x} \in \mathbb{R}^n$ of the dynamical system $\dot{\mathbf{x}}(\tau) = f(t; \mathbf{x}, \mathbf{u}, \mathbf{w})$, influenced by a pursuing
 84 player $\mathbf{P}$ (with control $\mathbf{w} \in \mathcal{W} \subseteq \mathbb{R}^p$) and its evading pair $\mathbf{E}$ (with control $\mathbf{u} \in \mathcal{U} \subseteq \mathbb{R}^m$), results in
 85 system trajectories $\xi_{\mathbf{x},\tau}^{\mathbf{u},\mathbf{w}}$ that are obtained by optimizing a scalar-valued payoff functional, $\ell(\xi_{\mathbf{x},t}^{\mathbf{u},\mathbf{w}}(\tau))$
 86 i.e.

$$\mathcal{J}(\tau; \mathbf{x}, \mathbf{u}, \mathbf{w}) = \min_{\tau \in [t, T]} \ell(\xi_{\mathbf{x},t}^{\mathbf{u},\mathbf{w}}(\tau)) \quad (\text{Payoff functional})$$

87 within a two-person differential game for a fixed $T \geq \tau > t$, where ℓ can be a signed distance to the
 88 (boundary of the) target set/region, for example. The controllers belong in compact sets which are
 89 measurable functions i.e. $\bar{\mathcal{U}} \equiv \mathbf{u} : [t, T] \rightarrow \mathcal{U}$, $\bar{\mathcal{W}} \equiv \mathbf{w} : [t, T] \rightarrow \mathcal{W}$.

90 2.2 A Differential Game with Two Inputs

91 This section provides a cursory background that is suitable for a minimal problem construction.
 92 Readers interested in a more detailed background can consult the appendix A.

93 For a (possibly infinite) *terminal time* T , a constant k , $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{W}$ and all $\hat{\mathbf{x}}$, $\mathbf{x} \in \mathbb{R}^n$, let the
 94 system possess a BUC payoff functional $\mathbf{g}(\mathbf{x}(T))$ that satisfies $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}$. The differential game's
 95 *lower value* is Evans and Souganidis [1984b]

$$\mathbf{v}(\mathbf{x}, t) = \inf_{\beta \in \mathcal{B}(t)} \sup_{\mathbf{u} \in \mathcal{U}(t)} \mathcal{J}(t; \mathbf{x}, \mathbf{u}, \mathbf{w}) \triangleq \inf_{\beta \in \mathcal{B}(t)} \sup_{\mathbf{u} \in \mathcal{U}(t)} \min_{\tau \in [t, T]} \ell(\xi_{\mathbf{x},t}^{\mathbf{u},\mathbf{w}}(\tau)). \quad (2)$$

96 is the viscosity solution to (HJ-IVP) Evans and Souganidis [1984b, Lemma 2.1] with Hamiltonian,

$$\mathbf{H}(t, \mathbf{x}, \mathbf{u}, \mathbf{w}, p) = \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \langle f(t, \mathbf{x}, \mathbf{u}, \mathbf{w}), p \rangle. \quad (3)$$

97 where the the pursuer $\mathbf{P}$ possesses a non-anticipative strategy $\mathcal{B} : \mathcal{W}(t) \rightarrow \mathcal{U}(t)$, beginning at
 98 time t , for each $t \leq \tau \leq T$ and $\mathbf{w}, \hat{\mathbf{w}} \in \mathcal{W}(t)$ such that $\mathbf{w}(\tau) = \hat{\mathbf{w}}(\tau)$ for $t \leq \tau < T$ almost
 99 everywhere. This implies that $\mathcal{B}[\mathbf{w}](\tau) = \mathcal{B}[\hat{\mathbf{w}}](\tau)$ for $t \leq \tau < T$ almost everywhere Evans and
 100 Souganidis [1984b]. The solution $\mathbf{v}(t; \mathbf{x})$ in (2) is the dynamic programming optimality condition
 101 for a differential game with two inputs in continuous time. It is popularly employed in constructing
 102 safe sets in backward reachability settings Mitchell et al. [2005].

103 2.3 Robustly Controlled Backward Reachable Set and Tube

104 Suppose that the goal of $\mathbf{E}$ is to reach a user-specified target region $\mathcal{L}_0(\tau)$ within $\tau \leq T$ time steps
 105 of playing the game; while the $\mathbf{P}$ simultaneously seeks to prevent this from happening. The invariant
 106 set $\mathcal{L}_0$ obtained at T

$$\mathcal{L}_0(T) = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{g}(0; \mathbf{x}) \leq 0\} \quad (\text{Target-Set})$$

107 is “*robustly controlled*” for the “*distance-to-target-set*” cost $\mathbf{g}(0; \mathbf{x})$, with constraints $|\mathbf{g}(0; \mathbf{x})| \leq$
 108 k , $|\mathbf{g}(0; \mathbf{x}) - \mathbf{g}(t; \hat{\mathbf{x}})| \leq k|\mathbf{x} - \hat{\mathbf{x}}|$ where $k > 0$ is a constant, and all $-T \leq t \leq 0$, $\{\hat{\mathbf{x}}, \mathbf{x} \in \mathbb{R}^n\}^1$.
 109 The distance to $\mathcal{L}_0(\tau)$ is typically found by optimizing $\ell(\mathbf{x}, t)$ as in (Payoff functional). The HJI
 110 PDE (2) and Hamiltonian (3) for this set admit the form

$$\mathbf{v}_t(t; \mathbf{x}) + \min\{0, \mathbf{H}(\mathbf{x}, \partial \mathbf{v}_{\mathbf{x}}(t; \mathbf{x}))\} = 0, \quad \mathbf{v}(0; \mathbf{x}) = \mathbf{g}(0; \mathbf{x}). \quad (\text{HJI-RCBRT})$$

111 For the safety problem setup in (2), the corresponding *robustly controlled backward reachable tube*
 112 (RCBRT) on $(0, T]$ is the closure of the open set Mitchell [2020]

$$\mathcal{L}([\tau, 0], \mathcal{L}_0) = \{\mathbf{x} \in \mathbb{R}^n \mid \exists \beta \in \bar{\mathcal{W}}(t) \forall \mathbf{u} \in \mathcal{U}(t), \exists \quad \tau \in [-T, 0], \xi(\bar{t}) \in \mathcal{L}_0\}. \quad (\text{Target-Tube})$$

113 Read: The set of states from which the strategies of $\mathbf{P}$ and for all controls of $\mathbf{E}$ imply that we *reach*
 114 *and remain in the target set* in the interval $[-T, 0]$. Following Lemma 2 of Mitchell et al. [2005], the
 115 states in the reachable set admit the following properties w.r.t the value function $\mathbf{v}^-$:

$$\mathbf{x}(t) \in \mathcal{L}(\cdot) \implies \mathbf{v}(\mathbf{x}, t) \leq 0, \quad \mathbf{v}(\mathbf{x}, t) \leq 0 \implies \mathbf{x}(t) \in \mathcal{L}(\cdot). \quad (\text{Zero Levelset})$$

¹Time is reversed in BRT computational scenarios.

116 In backward reach avoid tubes where the agent must avoid the unsafe region at all times, we can
 117 write (HJI-RCBRT) as the HJI variational inequality for a *robustly controlled backward reach-avoid*
 118 *tube* (RCBRAT) as follows,

$$\min\{v_t(t; \mathbf{x}) + \mathbf{H}(t; \mathbf{x}, \partial v_{\mathbf{x}}), g(\mathbf{x}, t) - v(t; \mathbf{x})\} \leq 0, \quad v(0; \mathbf{x}) = g(0; \mathbf{x}). \quad (\text{HJI-RCBRAT})$$

119 **Challenges in optimizing the RCBRT/RCBRAT:** Computing the RCBRT presents a host of chal-
 120 lenges for optimization because

- 121 • the dynamics $f(\cdot)$ and the nonlinear PDE are usually nonlinear with coupled states in most
 122 problems of interest;
- 123 • although $\mathcal{L}([\tau, 0], \mathcal{L}_0)$ is mostly smooth, it possesses discontinuous “shocks” at solution
 124 crossover points (see Mitchell [2020]); this makes global gradient-based optimization
 125 impossible for resolving v ;
- 126 • more than one trajectory can emanate from the RCBRT’s boundary that end up at the
 127 same point on the target set – this makes sampling-based optimization difficult, but not
 128 insurmountable; and
- 129 • the RCBRT is nonconvex.

130 To mitigate these challenges above, we introduce a set of tools that allow us to circumvent these
 131 challenges in an optimization context.

132 3 Methods

133 We first transform the nonlinear hyperbolic “smoothed” HJ PDE (HJ-Visc) to a linearized form and
 134 exact its unique solution based on a deduction from the solution to the heat equation. We then
 135 prescribe a sampling-based scheme, based on proximity theory, for its spatial gradients. This is found
 136 in a stochastic gradient descent algorithm in a zeroth-order optimization HJ scheme. The latter parts
 137 of this section extend the results to (HJI-RCBRT) and (HJI-RCBRAT).

138 3.1 HJ Linearization by a Cole-Hopf Transformation

139 Here, we first transform the smooth HJ equation (HJ-Visc) into a linear form via the Cole-Hopf
 140 trick Evans [2022, §4.4.1]. The Cole-Hopf transformation that linearizes (HJ-Visc) can be written as
 141 (see the derivation in Appendix B)

$$\varphi^\delta = \exp(-2v^\delta/\delta). \quad (\text{Cole-Hopf})$$

142 Applying the Cole-Hopf transformation to (HJ-Visc), we find that the fundamental solution to the
 143 smoothed version of the Hamilton-Jacobi equation (HJ-Visc) is given by (4).

144 **Lemma 1** (Smoothed HJ Solution). *The unique solution to the smoothed version of the Hamilton-*
 145 *Jacobi equation (HJ-Visc) is*

$$v^\delta(t; \mathbf{x}) = -\frac{\delta}{2} \log \left\{ (2\pi\delta t)^{-n/2} \int_{\mathbb{R}^n} \exp\left(-\frac{1}{2\delta t} \cdot |\mathbf{x} - \mathbf{y}|^2\right) \exp\left(-\frac{2}{\delta} \cdot v^\delta(\mathbf{y})\right) d\mathbf{y} \right\} \quad (4)$$

146 *in $\mathbb{R}^n \times (0, T]$ and $v^\delta(t; \mathbf{x}) = 0$ otherwise.*

147 **Corollary 1** (HJ solution as a log expectation). *On the domain of the differential equation, the unique*
 148 *solution can be written as the following expectation*

$$v^\delta(t; \mathbf{x}) = -\frac{\delta}{2} \log \mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} \left[\exp\left(-\frac{2}{\delta} v^\delta(\mathbf{y})\right) \right]. \quad (5)$$

149 *where $\mathbf{y}$ is drawn from a Gaussian distribution with mean $\mathbf{x}$ and variance δt .*

150 Inspecting (Cole-Hopf), we may write

$$v^\delta(t; \mathbf{x}) = -\frac{\delta}{2} \log (\varphi^\delta(t; \mathbf{x})) (\mathbf{x}) \text{ in } \mathbb{R}^n \times (0, T], \quad (6a)$$

151 from which we can write

$$\partial_{\mathbf{x}} \mathbf{v}^\delta(t; \mathbf{x}) = -\frac{\delta}{2} \cdot \frac{\nabla \varphi^\delta(t; \mathbf{x})}{\varphi^\delta(t; \mathbf{x})}. \quad (7)$$

152 Similarly, for $\mathbf{v}_t$, recall that $\varphi_t^\delta = \Omega'(\mathbf{v}^\delta) \mathbf{v}_t^\delta$, where $\Omega' = -(2/\delta) \exp(-\frac{2}{\delta} \mathbf{v}^\delta)$ is the derivative
153 obtained from the linear transformation (B.4). It follows that

$$\mathbf{v}_t^\delta = \frac{\varphi_t^\delta}{\Omega'(\mathbf{v}^\delta)} := \frac{\frac{\delta}{2} \Delta \varphi^\delta}{-(2/\delta) \exp(-\frac{2}{\delta} \mathbf{v}^\delta)} \quad (8)$$

154 from which we can write the time derivative as

$$\mathbf{v}_t^\delta = -\frac{\delta^2}{4} \frac{\Delta \varphi^\delta}{\exp(-\frac{2}{\delta} \mathbf{v}^\delta)}. \quad (9)$$

155 We now cast the computation of the optimal “controllable”² reachable set into a form that leverages
156 the Moreau envelope (Moreau-Env) in a successive approximation scheme so that the reachable set
157 can be obtained via the subdifferential operator of the problem.

158 3.2 Operators and Envelopes of HJ Equations

159 Monotone operator theory has had a huge impact on partial differential equations and nonlinear
160 systems Zarantonello [1971], Bot and Hendrich [2014], reducing nonlinear analyses to problems of
161 the form

$$\text{determine } \mathbf{x} \in \text{zer } \mathbf{Q} = \{\mathbf{x} \in \mathcal{H} \mid 0 \in \mathbf{Q}\mathbf{x}\}. \quad (10)$$

162 The subdifferential operator (see Bauschke et al. [2017, §16]), $\partial \ell : \mathcal{H} \rightarrow 2^{\mathcal{H}}$, of $g(\mathbf{x})$ at $\bar{\mathbf{x}}$ is the set
163 of all ϑ that satisfies

$$\{\vartheta \in \mathcal{H} \mid (\forall \mathbf{x} \in \mathcal{H}) \langle \mathbf{x} - \bar{\mathbf{x}}, \vartheta \rangle + \ell(\bar{\mathbf{x}}) \leq \ell(\mathbf{x})\}. \quad (11)$$

164 That is, ℓ is subdifferentiable at $\mathbf{x}$ if $\partial \ell(\mathbf{x}) \neq \emptyset$; hence, elements of $\partial \ell(\mathbf{x})$ are the subgradients of ℓ
165 at $\mathbf{x}$.

166 An optimization implication of (11) is Fermat’s rule, which states that for $\ell : \mathcal{H} \rightarrow (0, T)$ Moreau
167 [1965],

$$\text{argmin } \ell = \text{zer } \partial \ell. \quad (12)$$

168 The proximity operator of $\ell(\mathbf{x})$ at a time, $t \in (0, T]$ is the minimizers set Moreau [1962a]

$$\text{Prox}_{t\ell}(\mathbf{x}) \triangleq \arg \min_{\mathbf{z} \in \mathcal{H}} \left(\ell(\mathbf{z}) + \frac{1}{2t} \|\mathbf{z} - \mathbf{x}\|^2 \right) \quad (\text{Prox-Op})$$

169 that define the Moreau envelope $v(\cdot, t)$, defined as Moreau [1962b]

$$v(\mathbf{x}, t) = \min_{\mathbf{z} \in \mathcal{H}} \left(\ell(\mathbf{z}) + \frac{1}{2t} \|\mathbf{z} - \mathbf{x}\|^2 \right). \quad (\text{Moreau-Env})$$

170 The envelope v expands troughs of $\ell(\mathbf{x})$ and its local minimizers concur with those of ℓ Chaudhari
171 et al. [2018].

172 **Lemma 2.** Suppose that $\ell(\mathbf{x})$ is BUC, its subdifferential exists, the set $S_\alpha \triangleq \{\mathbf{z} \in \mathbb{R}^n : \ell(\mathbf{z}) \leq$
173 $\inf \ell + \alpha\}$ is compact, $\text{zer } \partial \ell$ exists and $\mathbf{x} \in S_\alpha$. An important result established in Heaton et al.
174 [2024, Lemma A.3] relates the spatial gradients of the solution to (HJ-IVP) to the proximity operator
175 of $\ell(\mathbf{x})$ at time t as

$$\text{Prox}_{t\ell}(\mathbf{x}) = \mathbf{x} - t \partial_{\mathbf{x}} v(t; \mathbf{x}) \approx \mathbf{x} - t \cdot \partial_{\mathbf{x}} \mathbf{v}^\delta(t; \mathbf{x}). \quad (13)$$

²We use this term because the agents in the differential game system have control inputs

176 3.3 HJ PDE's spatial gradients and its proximity operator

177 Consider the kernel $\varphi^\delta(\mathbf{x}, t)$ in (??) which corresponds to the solution of (HJ-Visc). Resolving the
 178 convolution (??) is intractable for large n and time step t . However, it is same as the probability
 179 density of a Gaussian distribution with mean $\mathbf{x}$ and variance δt (as derived in Heaton et al. [2024,
 180 Appendix B]), i.e.

$$\partial \varphi_{\mathbf{x}}^\delta(t; \mathbf{x}) = -\frac{1}{\delta t} \cdot \mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [(\mathbf{x} - \mathbf{y}) \exp(\ell(\mathbf{y})/\delta)]. \quad (14)$$

181 Plugging (??) and (14) into (7), we find that

$$t \cdot \partial \mathbf{v}_{\mathbf{x}}^\delta(t; \mathbf{x}) = \mathbf{x} - \frac{\mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\mathbf{y} \exp(-\ell(\mathbf{y})/\delta)]}{\mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\exp(-\delta^{-1} \ell(\mathbf{y}))]} \quad (15)$$

182 provides the time-scaled spatial gradients of $\mathbf{v}(t; \mathbf{x})$ in (18) without the need for unstable schemes
 183 employed in upwinding methods within grid discretization numerical reachable sets' computational
 184 algorithms ?Molu [2025], Mitchell [2004].

185 Substituting the r.h.s of (15) into (13), we find that

Proximal Operator for $\partial_{\mathbf{x}} \mathbf{v}^\delta(t; \mathbf{x})$

$$\text{Prox}_{t\ell}(\mathbf{x}) = \frac{\mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\mathbf{y} \cdot \exp(-\ell(\mathbf{y})/\delta)]}{\mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\exp(-\delta^{-1} \ell(\mathbf{y}))]} \quad (16)$$

187 3.4 Hamilton-Jacobi-Proximity Operator Relation

188 Now consider the HJI PDEs (HJI-RCBRT) and (HJI-RCBRAT) in light of (16). We can write the
 189 domain of these PDEs in light of the proximal expression (16) as

$$\begin{aligned} \partial \mathbf{v}_t(t; \mathbf{x}) + \min\{0, \mathbf{H}(t; \mathbf{x}, \partial \mathbf{v}_{\mathbf{x}}(t; \mathbf{x}))\} &= 0, \\ \approx \partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \min\{0, \mathbf{H}^\delta(t; \mathbf{x}, \partial \mathbf{v}_{\mathbf{x}}^\delta(t; \mathbf{x}))\} &= 0, \\ := \partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \min \left\{ 0, \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \left\langle f(t; \mathbf{x}, \mathbf{u}, \mathbf{w}), \frac{1}{t}(\mathbf{x} - \text{Prox}_{t\ell}(\mathbf{x})) \right\rangle \right\} &= 0, \end{aligned} \quad (17)$$

190 for (HJI-RCBRT). Similarly, the right-hand-side of the variational inequality for (HJI-RCBRAT) can
 191 be written as

$$\begin{aligned} \min\{\partial \mathbf{v}_t(t; \mathbf{x}) + \mathbf{H}(t; \mathbf{x}, \partial \mathbf{v}_{\mathbf{x}}(t; \mathbf{x})), \mathbf{g}(\mathbf{x}, t) - \mathbf{v}(t; \mathbf{x})\}, \\ \approx \min\{\partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \mathbf{H}^\delta(t; \mathbf{x}, \partial \mathbf{v}_{\mathbf{x}}^\delta(t; \mathbf{x})), \mathbf{g}(\mathbf{x}, t) - \mathbf{v}^\delta(t; \mathbf{x})\}, \\ := \min\{\partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \langle f(t; \mathbf{x}, \mathbf{u}, \mathbf{w}), \partial \mathbf{v}_{\mathbf{x}}^\delta(t; \mathbf{x}) \rangle, \mathbf{g}(t; \mathbf{x}) - \mathbf{v}^\delta(t; \mathbf{x})\}. \end{aligned} \quad (18)$$

192 Thus, evaluating the spatial gradient of $\mathbf{v}^\delta$ in (18) amounts to drawing samples using the proximity
 193 expression above and using (??) to evaluate $\mathbf{v}^\delta$. We can now write (HJI-RCBRAT) as

$$\min \left\{ \partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \left\langle f(t; \mathbf{x}, \mathbf{u}, \mathbf{w}), \frac{1}{t}(\mathbf{x} - \text{Prox}_{t\ell}(\mathbf{x})) \right\rangle, \right. \quad (19)$$

$$\left. \mathbf{g}(\mathbf{x}, t) + \delta \log \mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\exp(-\delta^{-1} \ell(\mathbf{y}))] \right\} \leq 0, \quad \mathbf{v}(0; \mathbf{x}) = \ell(0; \mathbf{x}). \quad (20)$$

194 3.5 Problem Algorithmics

195 We now cast the solution of the variational HJI computational problem as a Monte Carlo algorithmics.
 196 Towards finding global solutions to the optimization problem, we adapt the Hamilton-Jacobi Moreau
 197 adaptive descent algorithm of Heaton et al. [2024] to the computation of the HJI(HJI-RCBRT) and
 198 (HJI-RCBRAT).

199 The Moreau adaptive descent algorithm works by minimizing the Moreau envelope (Moreau-Env) of
 200 the HJ PDE in a nonconvex optimization scheme via a zeroth-order sampling scheme. In its native
 201 form, within a set convergence loop, it works by

Algorithm 1 Sampling-based RCBRT descent.

Given $\{\epsilon, \delta\} > 0$, and initial step size α .
function **ProxReach**($\mathbf{x}, \ell, \eta, t_0, T$) ▷ Sample size, η .
 Set $\tau = t_0, t_s = (T - t_0)/k$.
 while $T - \tau > \epsilon T$ **do**
 $\mathbf{y}[\tau] \leftarrow \mathcal{N}(\mathbf{x}, \delta)$; ▷ Draw finite samples from $\mathbf{x}$.
 $\mathbf{z}[\tau] \leftarrow \ell(\mathbf{y}[\tau])$; ▷ See equation (6).
 $\partial_{\mathbf{z}} \mathbf{v}(\tau; \mathbf{z}) \leftarrow \text{Prox}_{\tau \ell}(\mathbf{z}, \delta)$ ▷ Eq. (Prox-Op).
 $\tau \leftarrow \tau + t_s$
 $\mathbf{z}[\tau + t_s] \leftarrow \mathbf{z}[\tau] - \alpha \cdot \ell[\tau]$;
 $\tau \leftarrow \text{Adjuster}(\tau, \ell[\tau], \ell[\tau - \Delta t])$;
 end while
end function

- 202 • first computing the spatial gradient of $\mathbf{v}^\delta(t; \mathbf{x})$; then
- 203 • updating the states of the dynamic system in a gradient descent scheme (under an appropriate
- 204 step size); and lastly
- 205 • uses an adaptive forward-backward time-update scheme to update the index of time.

206 The algorithm is succinctly described in Algorithm 1. Given a large initial step size α , time interval
207 $(t_0, T]$ and the state values for a given problem $\mathbf{x}$, the algorithm proceeds in a reactive sampling-based
208 gradient descent fashion by drawing finite samples of size η from the (potentially large) state space
209 $\mathbf{x} \in \mathbb{R}^n$. The samples on Line 5 of Algorithm 1 are draw from a multinomial Gaussian filter for
210 a total of η samples whose mean and variance are $\mathbf{x}$ and δt , respectively in every iteration of the
211 algorithm. Within line 7 of Algorithm 1, the Hamiltonian for the system is computed but negated
212 to make the computational scheme conform tot he backward reachable setting (since the iterative
213 loop proceeds forward in the index of time in Algorithm 1). In all our implementations, the time
214 index within the loop is adaptively adjusted (i.e. Line 9) based on a version of the Armijo-Goldstein
215 condition Armijo [1966] to ensure that an appropriate time step is taken in every iteration as we seek
216 the minimum value of the Moreau envelope.

217 After convergence, the solution(s) $\{\mathbf{x}\}$ that minimize the value function are returned. We now
218 adopt the HJ-MAD algorithm to the resolution of HJ problems with adversarial control inputs in a
219 two-player differential game. This is delineated in Algorithm 1. After the optimization of the extrema
220 of the value function, finding the safe set is equivalent to executing a search algorithm within the
221 optimized value function to evaluate where the condition (Zero Levelset) are fulfilled. In the next
222 section, we will connect the scheme we have presented so far to the safe reachability resolution
223 problem of two rockets on a cartesian grid as originally used in our previous paper ?.

224 3.6 Convergence analysis of the algorithm

225 [TO-DO: Haoxiang, please follow the results in this link:](#)

226 4 Numerical Results

227 In this section, we provide a representative problem and amend it to the viscous HJI
228 PDE (HJI-RCBRT) to test and verify our hypothesis.

229 4.1 The Rockets Launch Problem

230 We consider the rocket launch problem of Dreyfus Dreyfus [1966] which was amended to a differential
231 game between two identical rockets, $\mathbf{P}$ and $\mathbf{E}$, on an (x, z) cross-section of a Cartesian plane in ?Molu
232 [2025]. We want to compute the usable part Isaacs [1999] or RCBRT Mitchell et al. [2005] of the
233 *approximate* terminal surface’s boundary for a predefined target set over a target tube. The usable part
234 entails the regions of the state-space for which the min-max operation over either strategy of $\mathbf{P}$ and
235 $\mathbf{E}$ is below zero and where the boundary of the usable part (BUP) constitute where (HJI-RCBRAT)
236 is exactly zero.

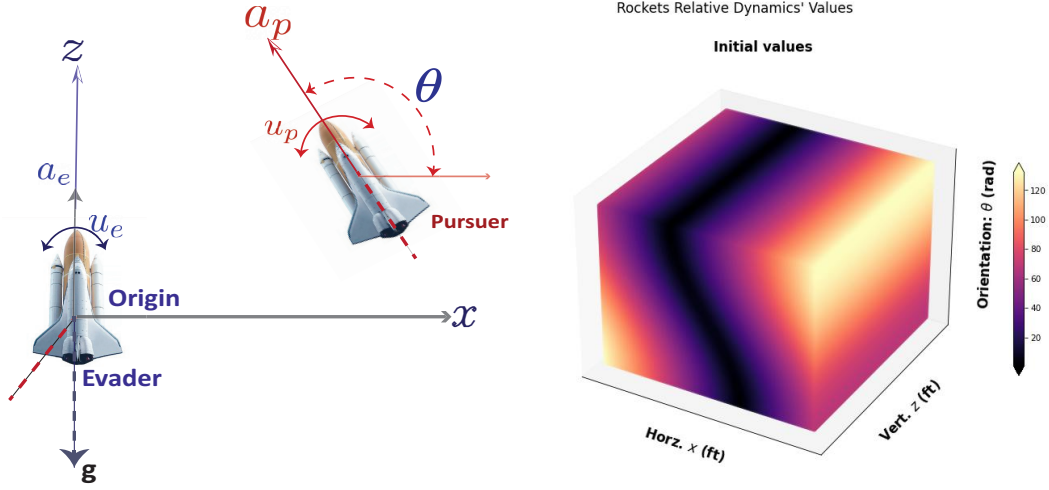


Figure 1: *Left*: Motion of two rockets on a Cartesian xz -plane with a thrust inclination in relative coordinates given by $\theta := u_p - u_e$. *Right*: Representation of the initial values as a heat map for the two rockets described in (26).

For a two-player differential game, let P and E share identical dynamics in a general sense so that we can freely choose the coordinates of P ; however, E 's origin is a distance Φ away from (x, z) at plane's origin (see Fig. 4.1) so that the PE vector's inclination measured counterclockwise from the x axis is θ .

Hence, we may consider the current problem as a terminal pursuit-evasion *differential* game, whereupon the game terminates when *capture* occurs. This eventual terminal represents the final *overapproximated* (non-)capture surface for the $P - E$ game (to be shortly introduced). We desire that every $\max$ and $\min$ operations during a *game of kind* are in the interior of the plane where motion exists; this way, no sudden changes from extremes are too aggravating in cost.

Let the states of P and E be (x_p, x_e) , respectively, driven by thrusts (u_p, u_e) respectively (see Figure 4.1). Fix the rockets' range so that what is left of the motion of either P or E 's is restricted to orientation on the (x, z) plane as illustrated in Fig. 4.1, where a and g are respectively the acceleration and gravitational accelerations (in feet per square second)³. As long as E remains within this target region or *backward reachable tube* (or BRT), P cannot cause damage or exercise an action of deleterious consequence on, say, the territory being guarded by E . Setting up E to maximize a payoff quantity with the largest possible margin or at least frustrate the efforts of P with minimal collateral damage while the pursuer minimizes this quantity constitutes a terminal value *optimal* differential game: there is no optimal pursuit without an optimal evasion.

The motion of P relative to E 's along the (x, z) plane includes the relative orientation, the control input, shown in Fig. 4.1 as $\theta = u_p - u_e$. Following the conventions in Fig. 4.1, the game's relative equations of motion in *reduced space* Isaacs [1999, §2.2] was derived as (see ?, §VI.A) is $\mathbf{x} = (x, z, \theta)$ where $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ and $(x, z) \in \mathbb{R}^2$ are

$$\dot{\mathbf{x}} \implies \{\dot{x} = a_p \cos \theta + u_e x, \dot{z} = a_p \sin \theta + a_e + u_e x - g, \dot{\theta} = u_p - u_e\} \quad (21)$$

The capture radius of the origin-centered circle Φ (we set $\ell = 1.5$ ft) is $\|\mathbf{PE}\|_2$ so that $\ell^2 = x^2 + z^2$. All capture points are specified by the variational HJ PDE Mitchell et al. [2005]:

$$\frac{\partial v}{\partial t}(\mathbf{x}, t) + \min \left[0, \mathbf{H} \left(\mathbf{x}, \frac{\partial \ell(\mathbf{x}, t)}{\partial \mathbf{x}} \right) \right] \leq 0, \quad (22)$$

with Hamiltonian given by

$$\mathbf{H}(\mathbf{x}, p) = - \max_{u_e \in [\underline{u}_e, \bar{u}_e]} \min_{u_p \in [\underline{u}_p, \bar{u}_p]} [p_1 \quad p_2 \quad p_3] \begin{bmatrix} a_p \cos \theta + u_e x \\ a_p \sin \theta + a_e + u_p x - g \\ u_p - u_e \end{bmatrix}. \quad (23)$$

³We set $a = 1 \text{ ft/sec}^2$ and $g = 32 \text{ ft/sec}^2$ in our simulation.

Here, p are the co-states, and $[\underline{u}_e, \bar{u}_e]$ denotes extremals that the evader must choose as input in response to the extremal controls that the pursuer plays i.e. $[\underline{u}_p, \bar{u}_p]$. Rather than resort to analytical *geometric reasoning*, we may analyze possibilities of behavior by either agent via a principled numerical simulation. This is the essence of this work. From (23), set $\underline{u}_e = \underline{u}_p = \underline{u} \triangleq -1$ and $\bar{u}_p = \bar{u}_e = \bar{u} \triangleq +1$ so that $\mathbf{H}(\mathbf{x}, p)$ is

$$- \max_{u_e \in [\underline{u}_e, \bar{u}_e]} \min_{u_p \in [\underline{u}_p, \bar{u}_p]} \left[\begin{array}{l} p_1(a_p \cos \theta + u_e x) + \\ p_2(a_p \sin \theta + a_e + \\ u_p x - g) + p_3(u_p - u_e) \end{array} \right] \\ \triangleq -ap_1 \cos \theta - p_2(g - a - a \sin \theta) - \bar{u}|p_1 x + p_3| + \underline{u}|p_2 x + p_3|, \quad (24)$$

where the last line follows from setting $a_e = a_p \triangleq a$. Thus, the rockets HJI equation (22) becomes

$$\partial_t v(t; \mathbf{x}) + \min [0, (-ap_1 \cos \theta - p_2(g - a - a \sin \theta) - |p_1 x + p_3| - |p_2 x + p_3|)] \leq 0, \quad (25)$$

where p_1, p_2, p_3 are the coefficients of the co-states obtained from (18). A similar derivation can be obtained for the (HJI-RCBRAT). The initial values of the game is as illustrated in ??.

4.2 Numerical Integration Considerations

We take the infinitesimal physics problem above and make it algorithmic first by discretizing the spatial and temporal domains i.e.

$$(x, z) \in (-100, +100], \theta \in (-\pi/2, \pi/2], t \in (0, 1], \quad (26)$$

with 1000 points along the temporal dimension and $\mathbf{x}_k - \mathbf{x}_{k-1} = 1/100$ along the spatial dimensions, where $k \in \mathbb{N}$. All our codes are implemented in pytorch, that runs on an Azure Linux node with the cpu processor specification: Intel Xeon Silver 4208 at 2.1GHz and 65GB memory; The pytorch runs on a Ubuntu 22.04 operating system.

We defined the target set as the top-left corner of the state space ?? and then run the Algorithm 1 on a time range of $(0, 1]$ seconds; we chose Dirichlet boundary conditions

$$\mathbf{v}^\delta(0; \mathbf{x}) = \mathbf{g}(0; \mathbf{x}) \triangleq 0. \quad (27)$$

The spatial gradients (co-states) were computed using the transformation (16) which were then used in generating the Hamiltonian and subsequently the solution to the safe set using (22).

5 Conclude

We have presented a gradient-based nonconvex optimization scheme for computing the safe set of a two-person differential game. The gradient scheme is a zero order Moreau envelope optimization scheme whose extrema coincides with that of the HJ-Isaacs equation (which is non-convex in its Hamiltonian). We provided a means to optimize the Proximal operator of the Moreau envelope by adapting the Heaton et al. [2024] algorithm and we note that once this extrema is found, determining the safe set amounts to searching within the resulting optimized terminal values of the HJI equation to see where the conditions of (Zero Levelset) are fulfilled. We demonstrated a proof of concept with the relative motion of two rockets on a Cartesian grid where both players have control inputs: the evading player must avoid the pursuing player within a pre-specified time range while the pursuing player must try to capture the evader within the given time by maximizing time. In a best response strategy, each player responds to the control input of its adversarial player in a best-response strategy. Our results demonstrate the feasibility of our approach.

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362 **A Background and Preliminaries (Unabridged).**

363 This section provides a context for the contribution of this work.

364 **A.1 Dynamic Programming and Two-Person Games.**

365 The formal relationships between the dynamic programming (DP) optimality condition for the *value*
366 in differential two-person zero-sum games, and the solutions to PDEs that solve “min-max” or
367 “max-min” type nonlinearity (the Isaacs’ equation) was presented in Isaacs [1999].

368 Essentially, Isaacs’ claim was that if the *value* functions are smooth enough, then they solve certain
369 first-order partial differential equations (PDE) problems with “max-min” or “min-max”-type nonlin-
370 earity. However, the DP value functions are seldom regular enough to admit a solution in the classical
371 sense.

372 “Weaker” solutions on the other hand Lions [1982], Evans and Souganidis [1984a], Crandall et al.
373 [1984], Crandall and Majda [1980b], Evans and Souganidis [1984b] provide generalized “viscosity”
374 solutions to HJ PDEs under relaxed regularity conditions; these viscosity solutions are not necessarily
375 differentiable anywhere in the state space, and the only regularity prerequisite in the definition is
376 continuity Crandall and Lions [1983]. However, wherever they are differentiable, they satisfy the
377 upper and lower values of HJ PDEs in a classical sense. Thus, they lend themselves well to many
378 real-world problems existing at the interface of discrete, continuous, and hybrid systems Lygeros
379 [2004], Osher and Sethian [1988], Mitchell [2020], Evans and Souganidis [1984b], Mitchell et al.
380 [2005].

381 Matter-of-factly, viscosity solutions to *Cauchy-type* HJ Equations admit usefulness in backward
382 reachability analysis Mitchell et al. [2005]. In scope and focus, this is the bulwark upon which we
383 build our formulation in this paper.

384 For a state $\mathbf{x} \in \Omega$ and a fixed time t : $0 \leq t < T$, suppose that the set of all controls for players $\mathbf{P}$
385 and $\mathbf{E}$ are respectively

$$\bar{\mathcal{U}} \equiv \{\mathbf{u} : [t, T] \rightarrow \mathcal{U} | \mathbf{u} \text{ measurable}, \mathcal{U} \in \mathbb{R}^m\}, \quad (\text{A.1})$$

$$\bar{\mathcal{W}} \equiv \{\mathbf{w} : [t, T] \rightarrow \mathcal{V} | \mathbf{w} \text{ measurable}, \mathcal{W} \subset \mathbb{R}^p\}. \quad (\text{A.2})$$

386 We are concerned with the differential equation,

$$\dot{\mathbf{x}}(\tau) = f(\tau, \mathbf{x}(\tau), \mathbf{u}(\tau), \mathbf{w}(\tau)), \quad \mathbf{x}(t) = \mathbf{x}, \quad T \leq \tau \leq t \quad (\text{A.3a})$$

where $f(\tau, \cdot, \cdot, \cdot)$ and $x(\cdot)$ are bounded and Lipschitz continuous. This bounded Lipschitz continuity property assures uniqueness of the system response $x(\cdot)$ to controls $u(\cdot)$ and $v(\cdot)$ Evans and Souganidis [1984b]. Associated with (A.3) is the payoff functional

$$\begin{aligned}\ell(u, w) &= \ell(t; x, u(\cdot), w(\cdot)) \\ &= \int_t^T l(\tau, x(\tau), u(\tau), w(\tau)) d\tau + g(x(T)),\end{aligned}\tag{A.4}$$

where $g(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies

$$|g(x)| \leq k_1, \quad |g(x) - g(\hat{x})| \leq k_1 |x - \hat{x}| \tag{A.5}$$

and $l : [0, T] \times \mathbb{R}^n \times \mathcal{U} \times \mathcal{W} \rightarrow \mathbb{R}$ is bounded and uniformly continuous, with

$$|l(t; x, u, w)| \leq k_2, \quad |l(t; x, u, w) - l(t; \hat{x}, u, w)| \leq k_2 |x - \hat{x}| \tag{A.6a}$$

for constants k_1, k_2 and all $0 \leq t \leq T$, $\hat{x}, x \in \mathbb{R}^n$, $u \in \mathcal{U}$ and $w \in \mathcal{W}$. We call T is the *terminal time* (it may be infinity!) and the integral, when it does not depend on the control laws, is the *performance index*. The evader's goal is to maximize the payoff (A.4) and pursuer's goal is to minimize it.

A.2 Upper and Lower Values of the Differential Game.

Suppose that the pursuer's mapping strategy (starting at t) is $\beta : \bar{\mathcal{U}}(t) \rightarrow \bar{\mathcal{W}}(t)$ provided for each $t \leq \tau \leq T$ and $u, \hat{u} \in \bar{\mathcal{U}}(t)$; then $u(\bar{t}) = \hat{u}(\bar{t})$ a.e. on $t \leq \bar{t} \leq \tau$ implies $\beta[u](\bar{t}) = \beta[\hat{u}](\bar{t})$ a.e. on $t \leq \bar{t} \leq \tau$. The differential game's lower value for a solution $x(t)$ that solves (A.3) for $u(t)$ and $v(t) = \beta[u](\cdot)$ is

$$\begin{aligned}v^-(t; x) &= \inf_{\beta \in \mathcal{B}(t)} \sup_{u \in \bar{\mathcal{U}}(t)} \ell(u, \beta[u]) \\ &= \inf_{\beta \in \mathcal{B}(t)} \sup_{u \in \bar{\mathcal{U}}(t)} \int_t^T l(\tau, x(\tau), u(\tau), \beta[u](\tau)) d\tau + g(x(T)).\end{aligned}\tag{A.7}$$

Similarly, suppose that the evader's mapping strategy (starting at t) is $\alpha : \bar{\mathcal{W}}(t) \rightarrow \bar{\mathcal{U}}(t)$ provided for each $t \leq \tau \leq T$ and $w, \hat{w} \in \bar{\mathcal{W}}(t)$; then $w(\bar{t}) = \hat{w}(\bar{t})$ a.e. on $t \leq \bar{t} \leq \tau$ implies $\alpha[w](\bar{t}) = \alpha[\hat{w}](\bar{t})$ a.e. on $t \leq \bar{t} \leq \tau$. The differential game's upper value for a solution $x(t)$ that solves (A.3) for $u(t) = \alpha[w](\cdot)$ and $w(t)$ is

$$\begin{aligned}v^+(t; x) &= \sup_{\alpha \in \mathcal{A}(t)} \inf_{w \in \bar{\mathcal{W}}(t)} \ell(\alpha[w], w) \\ &= \sup_{\alpha \in \mathcal{A}(t)} \inf_{w \in \bar{\mathcal{W}}(t)} \int_t^T l(\tau, x(\tau), \alpha[w](\tau), w(\tau)) d\tau + g(x(T)).\end{aligned}\tag{A.8}$$

These non-local PDEs i.e. (2) and (A.8) are hardly smooth throughout the state space so that they lack classical solutions even for smooth Hamiltonian and boundary conditions. However, these two values are "viscosity" (generalized) solutions Lions [1982], Crandall and Lions [1983] of the associated HJ-Isaacs (HJI) PDE, i.e. solutions which are *locally Lipschitz* in $\Omega \times [0, T]$, and with at most first-order partial derivatives in the Hamiltonian. In what follows, we introduce the notion of viscosity solutions to the HJI value functionals in (A.8), and (2).

A.3 Viscosity Solution of HJ-Isaac's Equations.

For any optimal control problem a value function is constructed based on the optimal cost (or payoff) of any input phase (x, T) . In reachability analysis, typically this is defined using a terminal cost function $g(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies

$$|g(x)| \leq k, \quad |g(x) - g(\hat{x})| \leq k |x - \hat{x}| \tag{A.9a}$$

for constant k and all $T \leq t \leq 0$, $\hat{x}, x \in \mathbb{R}^n$, $u \in \mathcal{U}$ and $w \in \mathcal{W}$. The zero sublevel set of $g(x)$ i.e.

$$\mathcal{L}_0 = \{x \in \bar{\Omega} \mid g(x) \leq 0\}, \tag{A.10}$$

416 **Lemma 3.** *The lower value v^- in (2) is the viscosity solution to the lower Isaac's equation*

$$\partial_t v^- + H^-(t; \mathbf{x}, \mathbf{u}, \mathbf{w}, \partial_{\mathbf{x}} v^-) = 0, \quad t \in [0, T], \quad \mathbf{x} \in \mathbb{R}^n \quad (\text{A.11a})$$

$$v^-(T; \mathbf{x}) = g(\mathbf{x}(T)), \quad \mathbf{x} \in \mathbb{R}^n \quad (\text{A.11b})$$

417 *with lower Hamiltonian,*

$$H^-(t; \mathbf{x}, \mathbf{u}, \mathbf{w}, p) = \max_{u \in \mathcal{U}} \min_{w \in \mathcal{W}} \langle \mathbf{f}(t; \mathbf{x}, \mathbf{u}, \mathbf{w}), p \rangle. \quad (\text{A.12})$$

418 *where p , the co-state, is the spatial derivative of v^- w.r.t $\mathbf{x}$.*

419 **Lemma 4.** *The upper value v^+ in (A.8) is the viscosity solution of the upper Isaac's equation*

$$\partial_t v^+ + H^+(t; \mathbf{x}, \mathbf{u}, \mathbf{w}, \partial_{\mathbf{x}} v^+) = 0, \quad t \in [0, T], \quad \mathbf{x} \in \mathbb{R}^n \quad (\text{A.13a})$$

$$v^+(T; \mathbf{x}) = g(\mathbf{x}(T)), \quad \mathbf{x} \in \mathbb{R}^n \quad (\text{A.13b})$$

420 *with upper Hamiltonian,*

$$H^+(t; \mathbf{x}, \mathbf{u}, \mathbf{w}, p) = \min_{w \in \mathcal{W}} \max_{u \in \mathcal{U}} \langle \mathbf{f}(t; \mathbf{x}, \mathbf{u}, \mathbf{w}), p \rangle, \quad (\text{A.14})$$

421 *with p being appropriately defined.*

422 **Corollary 2.** *(i) $v^- \leq v^+$ over $(t \in [0, T], \mathbf{x} \in \mathbb{R}^n)$ (ii) if for all $t \in [0, T], (\mathbf{x}, p) \in \mathbb{R}^n$, the*
 423 *minimax condition is satisfied i.e. $H^+(t; \mathbf{x}, \mathbf{u}, \mathbf{w}, p) = H^-(t; \mathbf{x}, \mathbf{u}, \mathbf{w}, p)$, then $v^- \equiv v^+$.*

424 **A.4 Reachability for Systems Verification.**

425 Reachability analysis is one of many verification methods that allows us to reason about (control-
 426 affine) dynamical systems. The verification problem may consist in finding a *set of reachable states*
 427 *that lie along the trajectory of the solution to a first order nonlinear partial differential equation that*
 428 *originates from some initial state $\mathbf{x}_0 = \mathbf{x}(0)$ up to a specified time bound, $t = t_f$. From a set of*
 429 *initial and unsafe state sets, the time-bounded safety verification problem is to determine if there is*
 430 *an initial state and a time within the bound that the solution to HJI PDE enters the unsafe set.*

431 Reachability could be analyzed in a

- 432 • *forward* sense, whereupon system trajectories are examined to determine if they enter certain
- 433 *states from an initial set;*
- 434 • *backward* sense, whereupon system trajectories are examined to determine if they enter
- 435 *certain target sets;*
- 436 • *reach set* sense, in which they are examined to see if states reach a set at a *particular time;*
- 437 *or*
- 438 • *reach tube* sense, in which they are evaluated that they reach a set at a point *during a time*
- 439 *interval.*

440 Backward reachability consists in avoiding an unsafe set of states under the worst-possible disturbance
 441 at all times; relying on nonanticipative control strategies. Backward reachable sets (BRS) and
 442 backward reachable tubes (BRTs) are popularly analyzed in a game of two vehicles with non-
 443 stochastic dynamics Merz [1972]. Such BRTs possess discontinuity at cross-over points (which
 444 exist at edges) on the surface of the tube, and may be non-convex. Therefore, treating the end-point
 445 constraints under these discontinuity characterizations need careful consideration and analysis when
 446 switching control laws if the underlying PDE does not have continuous partial derivatives (we discuss
 447 this further in section 3).

448 **A.5 Reachability from Differential Games Optimal Control**

449 For any admissible control-disturbance pair $(\mathbf{u}(\cdot), \mathbf{w}(\cdot))$ and initial phase $(\mathbf{x}_0, t_0)$, Crandall Crandall
 450 and Lions [1983] and Evan's Evans and Souganidis [1984a] claim is that there exists a unique function

$$\xi(t) = \xi(t; t_0, \mathbf{x}_0, \mathbf{u}(\cdot), \mathbf{w}(\cdot)) \quad (\text{A.15})$$

451 that satisfies (A.3) a.e. with the property that

$$\xi(t_0) = \xi(t_0; t_0, \mathbf{x}_0, \mathbf{u}(\cdot), \mathbf{w}(\cdot)) = \mathbf{x}_0. \quad (\text{A.16})$$

Read (A.15): the motion of (A.3) passing through phase (x_0, t_0) under the action of control u , and disturbance w , and observed at a time t afterwards. One way to design a system verification problem is compute the reachable set of states that lie along the trajectory (A.15) such that we evade the unsafe sets up to a time e.g. t_f within a given time bound e.g. $[t_0, t_f]$. In this regard, we discard the *cost-to-go*, $l(t; x(\tau), u(\tau), w(\tau))$ in (A.4), (2), or (A.8) and certify safety as resolving the terminal value, $g(x(T))$.

In backward reachability analysis, the lower value of the differential game is used in constructing an analysis of the backward reachable set (or tube). Therefore, we can cast a target set as the time-resolved terminal value $v^-(x, T) = g(x(T))$ so that given a time bound, and an unsafe set of states, the time-bounded safety verification problem consists in certifying that there is no phase within the target set (Target-Set) such that the solution to (A.3) enters the unsafe set. Following the backward reachability formulation of Mitchell et al. [2005], we say the zero sublevel set of $g(\cdot)$ in (A.11) i.e.

$$\mathcal{L}_0 = \{x \in \bar{\Omega} \mid g(x) \leq 0\}, \quad (\text{A.17})$$

is the *target set* in the phase space $\Omega \times \mathbb{R}$ for a backward reachability problem (proof in Mitchell et al. [2005]). This target set can represent the failure set, regions of danger, or obstacles to be avoided etc in the state space. Note that the target set, $\mathcal{L}_0$, is a closed subset of $\mathbb{R}^n$ and is in the closure of Ω . And the *robustly controlled backward reachable tube* for $\tau \in [-T, 0]^4$ is the closure of the open set

$$\mathcal{L}([\tau, 0], \mathcal{L}_0) = \{x \in \Omega \mid \exists \beta \in \bar{\mathcal{W}}(t) \forall u \in \mathcal{U}(t), \exists \bar{t} \in [-T, 0], \xi(\bar{t}) \in \mathcal{L}_0\}, \quad (\text{A.18})$$

Read: the set of states from which the strategies of P , and for all controls of E imply that we reach the target set within the interval $[T, 0]$. More specifically, following Lemma 2 of Mitchell et al. [2005], the states in the reachable set admit the following properties w.r.t the value function v

$$x \in \mathcal{L}_0 \implies v^-(x, t) \leq 0, \quad v^-(x, t) \leq 0 \implies x \in \mathcal{L}_0. \quad (\text{A.19a})$$

Observe:

- The goal of the pursuer, or P , is to drive the system's trajectories into the unsafe set i.e., P has u at will and aims to minimize the termination time of the game (c.f. (Target-Set));
- The evader, or E , seeks to avoid the unsafe set i.e., E has controls w at will and seeks to maximize the termination time of the game (c.f. (Target-Set));
- E has regular controls, u , drawn from a Lebesgue measurable set, $\mathcal{U}$ (c.f. (2)).
- P possesses *nonanticipative strategies* (c.f. (2)) i.e. $\beta[u](\cdot)$ such that for any of the ordinary controls, $u(\cdot) \in \mathcal{U}$ of E , P knows how to optimally respond to E 's inputs.

This is a classic reachability problem on the resolution of the infimum-supremum over the *strategies* of P and *controls* of E with the time of capture resolved as an extremum of a cost functional) over a time interval.

The terminal value $v^-(t; x)$, that characterizes the target set $\mathcal{L}_0$ is the viscosity solution to the HJI PDE

$$\frac{\partial v^-}{\partial t}(x, t) + \min\{0, H^-(t; x, u, w, v_x^-)\} = 0 \quad (\text{A.20a})$$

$$v^-(x, 0) = g(x), \quad (\text{A.20b})$$

where the vector field v_x^- is known in terms of the game's terminal conditions so that the overall game is akin to a two-point boundary-value problem. For ease of readability, we will remove the negative superscript on the lower value and Hamiltonian (3).

B Nonlinear HJI PDE Transformation

In this section, we construct the linearization and sampling machinery we use in building our results.

⁴The (backward) horizon, $-T$ is negative for $T > 0$.

490 B.1 Unique Solution to the viscous HJ Equation

491 Let us now construct the solution to the viscous initial-value-HJ problem.

492 *Proof of Lemma 1.* Recall from (HJ-Visc) that

$$v_t^\delta - \frac{\delta}{2} \Delta v^\delta + H(t; x, \partial_x v^\delta) = 0 \text{ in } \mathbb{R}^n \times (0, T], \quad v^\delta(0; x) = g(x) \text{ on } \mathbb{R}^n \times \{t = 0\}. \quad (\text{B.1})$$

493 Let $\varphi^\delta(t; x) = \Omega(v^\delta(t; x))$, where $\Omega : \mathbb{R}^n \rightarrow \mathbb{R}$ is a smooth function. Our goal is to choose Ω so
 494 that φ^δ solves a linear partial differential equation. Differentiating, we find that $\varphi_t^\delta = \Omega'(v^\delta) v_t^\delta$ and
 495 $\Delta \varphi^\delta = \Omega(v^\delta) \Delta v^\delta + \Omega''(v^\delta) H(t; x, \partial_x v^\delta)$. Thus, (HJ-Visc) implies

$$\varphi_t^\delta = \Omega' v_t^\delta \triangleq \Omega'(v^\delta) \left(\frac{\delta}{2} \Delta v^\delta - H(t; x, \partial_x v^\delta) \right) \quad (\text{B.2a})$$

$$= \frac{\delta}{2} \left(\Delta \varphi^\delta - \Omega''(v^\delta) H(t; x, \partial_x v^\delta) \right) - \Omega'(v^\delta) H(t; x, \partial_x v^\delta) \quad (\text{B.2b})$$

$$= \frac{\delta}{2} \Delta \varphi^\delta - \left(\frac{\delta}{2} \Omega''(v^\delta) + \Omega'(v^\delta) \right) H(t; x, \partial_x v^\delta). \quad (\text{B.2c})$$

496 We thus have

$$\varphi_t^\delta \triangleq \frac{\delta}{2} \Delta \varphi^\delta \quad (\text{B.3})$$

497 if we set the differential equation $\frac{\delta}{2} \Omega''(v^\delta) + \Omega'(v^\delta)$ to 0 — this is possible if we choose

$$\Omega(v^\delta) = \exp \left(-\frac{2\lambda}{\delta} \right). \quad (\text{B.4})$$

498 Whence, if v^δ solves the partial differential equation (HJ-Visc), we must have (Cole-Hopf) i.e.

$$\varphi^\delta = \exp(-2v^\delta/\delta). \quad (\text{B.5})$$

499 Equation (Cole-Hopf) is the so-called Cole-Hopf transformation Evans [2022, §4.4.1]; it serves as
 500 the solution to the initial-value problem for the heat equation (with conductivity $\delta/2$),

$$\begin{aligned} \partial_t \varphi^\delta(t; x) - \frac{\delta}{2} \Delta \varphi^\delta(t; x) &= 0 & \text{in } \mathbb{R}^n \times (0, T] \\ \varphi^\delta(t; x) &= \exp(-2v^\delta/\delta) & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{aligned} \quad (\text{Heat-Equation})$$

501 Whence, the unique bounded solution to (Cole-Hopf) is

$$\varphi^\delta(t; x) = (2\pi\delta t)^{(-n/2)} \int_{\mathbb{R}^n} \exp \left(-\frac{1}{2\delta t} |x - y|^2 \right) \exp \left(-\frac{2}{\delta} v^\delta(y) \right) dy, \quad x \in \mathbb{R}^n, t > 0. \quad (\text{B.6})$$

502 And since we can write $v^\delta = (\delta/2) \log \varphi^\delta$, it follows that the unique bounded solution to the
 503 initial-value (HJ-Visc) is (4), i.e.

$$v^\delta(t; x) = -\frac{\delta}{2} \log \left\{ (2\pi\delta t)^{-n/2} \int_{\mathbb{R}^n} \exp \left(-\frac{1}{2\delta t} |x - y|^2 \right) \exp \left(-\frac{2}{\delta} v^\delta(y) \right) dy \right\} \quad (\text{B.7})$$

504 □

505 *Proof of Corollary 1.* The proof is straightforward. Examining the solution (4), and reordering
 506 (Cole-Hopf), it is obvious that the right-hand-side is a form of the multivariate Gaussian distri-
 507 bution if we set the mean to x and the variance to δt . *A fortiori*, the solution to the HJ equation can
 508 be obtained by sampling from the multivariate Gaussian distribution (5). □

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